FORMULA SHEET · 01

PCL-AI Formula Sheet

PCI AI Project Controls Leader — the complete quantitative reference

CPI · SPI · EAC · TCPI · TF · EMV

PCI AI · PROJECT CONTROLS INSTITUTE GLOBAL
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AI proposes. The professional disposes.

How to use this sheet

Every formula the PCL-AI Body of Knowledge teaches, in one place, with the Knowledge Area where it is worked in full.

Three columns matter. **Formula** is the expression. **Meaning** is what it answers. **Watch for** is the mistake that formula actually attracts in practice — the reason a correct expression still produces a wrong answer.

This sheet is a study aid. It is **not provided in the examination**; candidates are expected to know the core formulas.

Conventions. Currency in USD unless stated. Ratios and indices to 2 dp. Adverse amounts in parentheses. Periods are consistent within any single calculation — never mix monthly and cumulative figures in one expression.

> **The notation clash rule.** `PV` means **Planned Value** in earned-value contexts and **present value** in discounting contexts. Write "present value" in words, or `PV(x)`, whenever discounting. These two collide on the same page more often than any other pair of symbols in the discipline.

1. Notation

SYMBOL	MEANING	UNIT
`A`, `L`, `E`	Assets, liabilities, equity	currency
`PV` (BCWS)	Planned value / budgeted cost of work scheduled	currency
`EV` (BCWP)	Earned value / budgeted cost of work performed	currency
`AC` (ACWP)	Actual cost / actual cost of work performed	currency
`BAC`	Budget at completion	currency
`CV`, `SV`	Cost variance, schedule variance	currency
`CPI`, `SPI`	Cost performance index, schedule performance index	ratio
`EAC`, `ETC`	Estimate at completion, estimate to complete	currency
`VAC`	Variance at completion	currency
`TCPI`	To-complete performance index	ratio
`ES`, `AT`	Earned schedule, actual time	periods
`PoC`	Percentage of completion	%
`O`, `M`, `P`	Optimistic, most likely, pessimistic duration	periods
`tE`, `σ`	PERT expected duration, standard deviation	periods
`ES`, `EF`, `LS`, `LF`	Early start, early finish, late start, late finish	date/period
`TF`, `FF`	Total float, free float	periods
`r`, `n`	Discount rate per period, number of periods	ratio, count
`EMV`	Expected monetary value	currency

2. Accounting and financial reporting · Domains 1–2

FORMULA	MEANING	WATCH FOR	KA
`A = L + E`	The accounting equation	—	1.1.1
`Σ Debits = Σ Credits`	Double-entry invariant	A trial balance that balances proves arithmetic, not correctness	1.1.3
`Retained earnings = Opening + Income − Expenses − Distributions`	Equity movement	—	1.1.1
`Depreciation = (Cost − Residual) ÷ Useful life`	Straight line	Residual omitted; life not reassessed	1.3.4
`Carrying amount = Cost − Accumulated depreciation − Impairment`	Net book value	—	1.3.4
`Expected value = Σ (probability × outcome)`	Provisions, variable consideration	Using most-likely outcome where the population is large	1.4.3 / 2.2.4
`Present value = Amount ÷ (1 + r)^n`	Discounting a provision	—	1.4.3
`PoC = Costs incurred ÷ Total estimated costs`	Cost-to-cost percentage of completion	Uninstalled materials inflating the numerator	2.2.6
`Cumulative revenue = PoC × Transaction price`	IFRS 15 over-time revenue	Applying to cumulative, then forgetting to deduct revenue already recognised	2.2.6
`Allocated price = Transaction price × (SSP_i ÷ Σ SSP)`	IFRS 15 allocation	—	2.2.5
`Contract asset / (liability) = Revenue recognised − Amounts billed`	Over/under-billing position	Sign convention reversed	7.5

3. Budgeting, cost and performance · Domains 3–5

FORMULA	MEANING	WATCH FOR	KA
`BAC = Σ control-account budgets + contingency`	Cost baseline	Management reserve wrongly included	3.1.4
`Total authorised budget = BAC + Management reserve`	Full authorisation	—	3.1.4
`Analogous estimate = Past cost × (this driver ÷ past driver)`	Analogous estimating	Driver not actually the cost driver	3.2.2
`Parametric estimate = Parameter × Rate`	Parametric estimating	Rate from a different scope basis	3.2.2
`Total cost = Fixed + (Variable per unit × Volume)`	Cost behaviour	—	5.1.1
`OAR = Budgeted overhead ÷ Budgeted activity base`	Overhead absorption rate	—	5.1.3
`Over/(under) absorption = Absorbed − Incurred`	Absorption variance	—	5.1.3
`Cost to date = Actuals + Accruals`	**True cost to date**	**The single most damaging omission in project controls**	5.2.1
`Price variance = (Actual price − Standard price) × Actual quantity`	Variance decomposition	Standard quantity used in place of actual	4.2.3
`Quantity variance = (Actual qty − Standard qty) × Standard price`	Variance decomposition	Actual price used in place of standard	4.2.3
`Total variance = Price variance + Quantity variance`	Reconciliation check	If the two do not sum to the total, one used the wrong multiplier	4.2.3

> **The accrual rule.** Every index below depends on `AC`. If `AC` excludes work performed but not yet invoiced, `CPI` overstates performance and `EAC` understates cost — and both errors are invisible in the report. Reconcile goods-received-not-invoiced to booked accruals before computing anything in Section 4.

4. Earned value and forecasting · Domains 6, 9

4.1 Position

FORMULA	MEANING	WATCH FOR	KA
`CV = EV − AC`	Cost variance	Negative is adverse	6.2.1
`SV = EV − PV`	Schedule variance (in currency)	Goes to zero at completion regardless of lateness	6.2.1
`CPI = EV ÷ AC`	Cost efficiency to date	Computed on invoiced cost only	6.2.2
`SPI = EV ÷ PV`	Schedule efficiency in cost terms	Also returns to 1.00 at completion	6.2.2
`% complete = EV ÷ BAC`	Progress by value	Not the same as physical or time progress	6.1

4.2 Forecast

FORMULA	ASSUMPTION IT ENCODES	KA
`EAC = AC + ETC`	Identity — always true	6.3.1
`EAC = AC + (BAC − EV)`	The variance was a **one-off**; remaining work performs to plan	6.3.2
`EAC = BAC ÷ CPI`	Cost performance to date **persists** to completion	6.3.2
`EAC = AC + (BAC − EV) ÷ CPI`	Same assumption, stated in remaining-work form	6.3.2
`EAC = AC + (BAC − EV) ÷ (CPI × SPI)`	**Both** cost and schedule pressure persist	6.3.2
`VAC = BAC − EAC`	Variance at completion; negative is an overrun	6.3.4
`TCPI (to BAC) = (BAC − EV) ÷ (BAC − AC)`	Efficiency required to still hit the budget	6.2.3
`TCPI (to EAC) = (BAC − EV) ÷ (EAC − AC)`	Efficiency required to hit a revised forecast	6.2.3

Identity worth knowing. `BAC ÷ CPI` is algebraically identical to `AC × BAC ÷ EV`. And when `EAC = BAC ÷ CPI`, then `TCPI (to EAC) = CPI` exactly — the forecast assumes you continue performing precisely as you have been. If a forecast produces a TCPI far above the CPI achieved to date, it is asserting an improvement nobody has yet demonstrated.

4.3 Earned schedule

FORMULA	MEANING	WATCH FOR	KA
`ES = M + (EV − PV_M) ÷ (PV_M+1 − PV_M)`	Interpolate between the cumulative-PV periods bracketing EV	Using period rather than cumulative PV	6.4.3
`SV(t) = ES − AT`	Schedule variance in **time**	—	6.4.3
`SPI(t) = ES ÷ AT`	Time-based schedule efficiency	—	6.4.3

Why earned schedule exists. `SV` and `SPI` are computed in currency and both converge to zero and 1.00 at completion, however late the project finishes. `SV(t)` and `SPI(t)` do not, which makes them the more honest late-stage measures.

4.4 Adaptive delivery

FORMULA	MEANING	WATCH FOR	KA
`% complete = Points completed ÷ Total planned points`	AgileEVM progress	Point inflation across sprints	9.5.3
`EV = % complete × BAC`	Earned value in adaptive delivery	—	9.5.3
`Velocity = Points completed ÷ Sprint`	Delivery rate	Averaging across a changed team	9.3
`Sprints remaining = Points remaining ÷ Velocity`	Forecast to completion	Using best-ever velocity rather than rolling average	9.3
`Run rate = Team cost ÷ Sprint`	Cost per sprint	—	9.5.2
`Cycle time = WIP ÷ Throughput`	Little's Law	Only valid for a stable system over the measured period	9.4

5. Scheduling and risk · Domains 10, 12

FORMULA	MEANING	WATCH FOR	KA
`EF = ES + Duration`	Forward pass	Off-by-one at period boundaries	10.2
`LS = LF − Duration`	Backward pass	—	10.2
`TF = LS − ES = LF − EF`	Total float	If the two expressions disagree, the pass is wrong	10.2.4
`FF = min(successor ES) − EF`	Free float	Ignoring lags on the successor link	10.2.4
`Critical path: TF = 0`	Longest path	Constraints creating artificial zero float	10.2.3
`tE = (O + 4M + P) ÷ 6`	PERT expected duration	Three points sourced from one person	10.1.4
`σ = (P − O) ÷ 6`	PERT standard deviation	—	10.1.4
`σ^2 path = Σ σ^2 of activities on the path`	Path variance	Variances add; standard deviations do not	10.3.4
`Crash cost slope = (Crash cost − Normal cost) ÷ (Normal duration − Crash duration)`	Cost per period saved	Crashing an activity that is not on the critical path	10.3.1
`EMV = Probability × Impact`	Expected monetary value	Applied to a one-off event where the expected value never occurs	12.2.3
`Contingency ≈ Σ EMV (or P80 − P50 from simulation)`	Risk-based contingency	Percentage applied in place of quantification	12.3.1

> **Variance adds, deviation does not.** To combine schedule uncertainty along a path, sum the **variances** (`σ^2`), then take the square root. Summing standard deviations overstates path uncertainty substantially.

6. Commercial · Domain 7

FORMULA	MEANING	WATCH FOR	KA
`Fee = Target fee + Share ratio × (Target cost − Actual cost)`	CPIF incentive fee	Sign convention on an overrun	7.1.3
`Pain/gain = Share ratio × (Actual − Target)`	Target-cost mechanism	—	7.1.4
`LD exposure = LD rate × Days late`	Liquidated damages	Cap not applied	7.2.3
`Amount due = Σ(% complete × item amount) − Retention − Previous payments`	Interim payment application	Retention release omitted at the milestone	7.4.3
`Retention released = Retention held × Release %`	Retention mechanics	—	7.4.4
`Remeasured value = Actual quantity × BoQ rate`	Remeasurement contract	Rate applied to the wrong measured unit	7.3.4

7. Process cycles and working capital · Domain 11

FORMULA	MEANING	WATCH FOR	KA
`DSO = Receivables ÷ Daily revenue`	Days sales outstanding	Revenue not annualised consistently	11.1.3
`DIO = Inventory ÷ Daily COGS`	Days inventory outstanding	—	11.A.1
`DPO = Payables ÷ Daily COGS`	Days payables outstanding	—	11.A.1
`CCC = DSO + DIO − DPO`	Cash conversion cycle	Sign on DPO	11.A.1
`Cash released ≈ Δ DSO × Daily revenue`	Working capital freed	—	11.A.1

8. Decision aid — which EAC method

The formula is not the difficulty. Selecting and defending the assumption is.

QUESTION	IF YES	METHOD
Was the variance caused by a discrete, closed event that will not recur?	→	`EAC = AC + (BAC − EV)`
Does the remaining work resemble the work done, with performance likely to persist?	→	`EAC = BAC ÷ CPI`
Is the project also behind, with schedule pressure driving cost?	→	`EAC = AC + (BAC − EV) ÷ (CPI × SPI)`
Is the remaining scope materially different from what has been done?	→	Bottom-up ETC; indices do not apply

Publish the range, not the point. Report at least two scenarios with the assumption named against each. A single EAC quoted to the nearest thousand implies a precision the method cannot support.

9. Decision aid — which progress measurement method

WORK PACKAGE CHARACTER	METHOD	NOTE
Short duration, spans one period	0/100	No partial credit; simplest and least gameable
Two periods, discrete start and finish	50/50	Credit on start and completion
Countable, homogeneous output	Units complete	Strongest objective measure
Long duration, distinct verifiable stages	Milestone weighting	Weights fixed **before** work starts
Level of effort, no discrete output	Apportioned / LoE	Earns with time; never the default
Cannot be measured physically	—	Re-scope the work package

> **The rule.** Fix the measurement method per work package before work begins, and never accept a percentage supplied by the party being measured. That figure is a commercial position, not a measurement.

10. Worked micro-examples

Each is internally consistent and can be reproduced from the sheet.

Earned value at a data date. `PV` 168,000 · `EV` 151,200 · `AC` invoiced 158,000 · accrual 12,400 · `BAC` 420,000.

- True `AC` = 158,000 + 12,400 = **170,400**
- `CPI` = 151,200 ÷ 170,400 = **0.887** (on invoiced cost alone it would read 0.957)
- `SPI` = 151,200 ÷ 168,000 = **0.900**
- `EAC` one-off = 170,400 + (420,000 − 151,200) = **439,200**
- `EAC` CPI persists = 420,000 ÷ 0.887 = **473,300**
- `EAC` both persist = 170,400 + 268,800 ÷ (0.887 × 0.900) = **507,000**
- `VAC` range = **(19,200)** to **(87,000)**

The accrual alone moves the forecast by 34,400 — the difference between a contained overrun and an uncontained one.

Earned schedule. Cumulative `PV` by month: 20, 55, 100, 150, 210, 280. `EV` = 170 at `AT` = 5.

- `EV` falls between month 4 (150) and month 5 (210)
- `ES` = 4 + (170 − 150) ÷ (210 − 150) = 4 + 0.333 = **4.333**
- `SV(t)` = 4.333 − 5 = **(0.667) months**
- `SPI(t)` = 4.333 ÷ 5 = **0.867**

PERT. `O` = 8, `M` = 12, `P` = 22.

- `tE` = (8 + 48 + 22) ÷ 6 = **13.0**
- `σ` = (22 − 8) ÷ 6 = **2.33**; `σ^2` = **5.44**

Float. `ES` = 4, duration 6, `LF` = 14.

- `EF` = 10 · `LS` = 8 · `TF` = 8 − 4 = **4** = 14 − 10 — the two forms agree

Working capital. Receivables 4,500,000 · daily revenue 50,000 · inventory 2,000,000 · daily COGS 40,000 · payables 3,200,000.

- `DSO` = **90** · `DIO` = **50** · `DPO` = **80** · `CCC` = 90 + 50 − 80 = **60 days**

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11. The ten formulas most often misapplied

1. **`CPI` computed on invoiced cost.** Without accruals, every downstream index and forecast is wrong in the same direction.
2. **`SV` read as a schedule measure late in a project.** It is denominated in currency and converges to zero at completion regardless of lateness. Use `SV(t)`.
3. **A single `EAC` presented as the forecast.** Three defensible answers exist. The range and the named assumption are the deliverable.
4. **`TCPI` ignored as a reality check.** A TCPI well above the CPI achieved to date asserts an unearned improvement.
5. **Percentage complete supplied by the measured party.** A commercial position presented as a measurement.
6. **Standard deviations summed along a schedule path.** Variances add; take the square root at the end.
7. **`EMV` applied to a single one-off event.** Expected value describes a portfolio of risks, not a binary outcome that will either happen or not.
8. **Contingency set as a percentage of budget.** Unless it reconciles to quantified risk exposure, it is a habit rather than an analysis.

9. **Crashing an activity off the critical path.** Cost incurred, zero duration saved.
10. **Management reserve included in `BAC`.** `BAC` is the cost baseline; management reserve sits outside it and is released by governance.

12. Cross-references

FOR THE FULL TREATMENT	SEE
Every formula worked with figures and self-check exercises	The PCL-AI Body of Knowledge, at the KA cited in each row
The published syllabus and weightings	PCL-AI Examination Content Outline
Why cut-off and accrual discipline dominate outcomes	Knowing Early, Knowledge Series 01, §4 and §16
Assessing whether a professional can defend these methods	Show Your Work, Knowledge Series 02, §4 and §8

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